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Date: 08/26/2024

Instruction

What the question is

Regular Expression Practice: 1.18

1.18 Give regular expressions generating the languages of Exercise 1.6.

1.6 Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is $\{0,1\}$.

a. $\{w \mid w$ begins with a 1 and ends with a 0$\}$
b. $\{w \mid w$ contains at least three 1s$\}$
c. $\{w \mid w$ contains the substring 0101 (i.e., $w = x0101y$ for some x and y)$\}$
d. $\{w \mid w$ has length at least 3 and its third symbol is a 0$\}$
e. $\{w \mid w$ starts with 0 and has odd length, or starts with 1 and has even length$\}$
f. $\{w \mid w$ doesn't contain the substring 110$\}$
g. $\{w \mid$ the length of w is at most 5$\}$
h. $\{w \mid w$ is any string except 11 and 111$\}$
i. $\{w \mid$ every odd position of w is a 1$\}$
j. $\{w \mid w$ contains at least two 0s and at most one 1$\}$
k. $\{\varepsilon, 0\}$
l. $\{w \mid w$ contains an even number of 0s, or contains exactly two 1s$\}$
m. The empty set
n. All strings except the empty string

1.18 1.6 Exercise A-n:

1.18a $\Sigma=\{0,1\}$ $R=1\Sigma^*0=1(0+1)^*0$

10, 100, 110, 1010, 1100, 10, 10100

$\boxed{1(0+1)^*0}$

1.18b R $\Sigma=\{0,1\}$ 111, 010101, 01101, 0000111, ...

$R=\Sigma^*1\Sigma^*1\Sigma^*1\Sigma^*$

$=(0+1)^*1(0+1)^*1(0+1)^*1(0+1)^*$

$\boxed{(0+1)^*1(0+1)^*1(0+1)^*1(0+1)^*}$

1.18c $\Sigma=\{0,1\}$ - This is gonna be for every single one, unless otherwise specified

$R=\Sigma^*0101\Sigma^*=(0+1)^*0101(0+1)^*$

0101, 001011, 101011, 1101010, ...

$\boxed{(0+1)^*0101(0+1)^*}$

1.18D $R=\Sigma\Sigma0\Sigma^*=(0+1)(0+1)0(0+1)^*$

000, 1101, ...

$\boxed{(0+1)(0+1)0(0+1)^*}$

1.18E $R=0(\Sigma\Sigma)^*+1\Sigma(\Sigma\Sigma)^*$

$=0((0+1)(0+1))^*+1(0+1)((0+1)(0+1))^*$

0, 011, 010, 00101, 10, 11, 1001

$\boxed{0((0+1)(0+1))^*+1(0+1)((0+1)(0+1))^*}$

1.18F $R=0^*(10^+)^*1^*$ 010, 011, 0101

$\boxed{0^*(10^+)^*1^*}$

1.18G $R=\varepsilon+\Sigma+\Sigma\Sigma+\Sigma\Sigma\Sigma+\Sigma\Sigma\Sigma\Sigma+\Sigma\Sigma\Sigma\Sigma\Sigma$

$=\varepsilon+(0+1)+(0+1)(0+1)+(0+1)(0+1)(0+1)+(0+1)(0+1)(0+1)(0+1)$

$+(0+1)^{\times 5}$ ε, 0, 01, 101, 10100000, ...

$=\varepsilon+(0+1)+(0+1)^2+(0+1)^3+(0+1)^4+(0+1)^5$ $\boxed{\varepsilon+(0+1)+(0+1)^2+(0+1)^3+(0+1)^4+(0+1)^5}$

1.18h $R = \varepsilon + \Sigma + 0\Sigma + 10 + 0\Sigma\Sigma + 10\Sigma + 110 + \Sigma^3\Sigma^+$

$\varepsilon + (0+1) + 0(0+1) + 10 + 0(0+1)(0+1) + 10(0+1) + 110 + (0+1)^3(0+1)$

$\varepsilon, 101, 110, 1010$

$\boxed{\varepsilon + (0+1) + 0(0+1) + 10 + 0(0+1)(0+1) + 10(0+1) + 110 + (0+1)^3(0+1)^+}$

1.18i $R = (1\Sigma)^*(\varepsilon+1) = (1(0+1))^*(\varepsilon+1)$

$\varepsilon, 101, 111, 1010$

$\boxed{(1(0+1))^*(\varepsilon+1)}$

1.18j $R = 00^*00^*(\varepsilon+1) + (00^*(\varepsilon+1)00^* + (\varepsilon+1)00^*00^*$

$001, 010, 100$

00^*

$00^*00^*(\varepsilon+1)$

$00^*00^*(\varepsilon+1) + 00^*(\varepsilon+1)00^* + (\varepsilon+1)00^*00^*$

1.18k $L = \{\varepsilon, 0\}$ $\Sigma = \{0, 1\}$

$R = 0 + \varepsilon$

$\boxed{0+\varepsilon}$

1.18l $R = 1^*(01^*01^*)^* + 0^*10^*10^*$

$\varepsilon, 00, 11, 0101, 010100 \ldots$

$\boxed{1^*(01^*01^*)^* + 0^*10^*10^*}$

1.18m L = The empty set

$R = \emptyset$

$\boxed{\emptyset}$

1.18 $R = \Sigma^* = (0+1)^+$

$\boxed{(0+1)^+}$

RE to NFA Practice: 1.19a, 1.19b

1.19 Use the procedure described in Lemma 1.55 to convert the following regular expressions to nondeterministic finite automata.

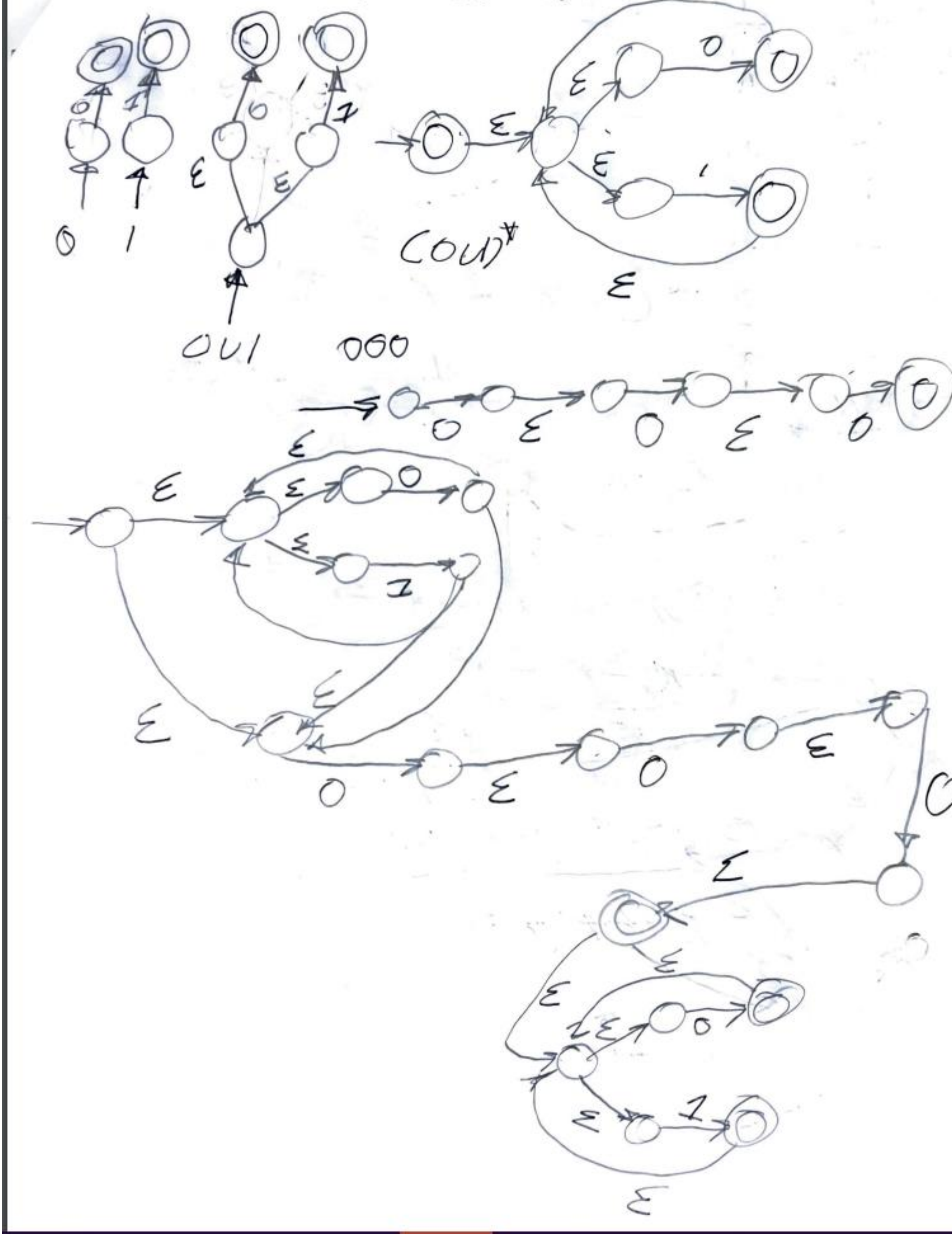
a. $(0 \cup 1)^*000(0 \cup 1)^*$

b. $(((00)^*(11)) \cup 01)^*$

[redacted]

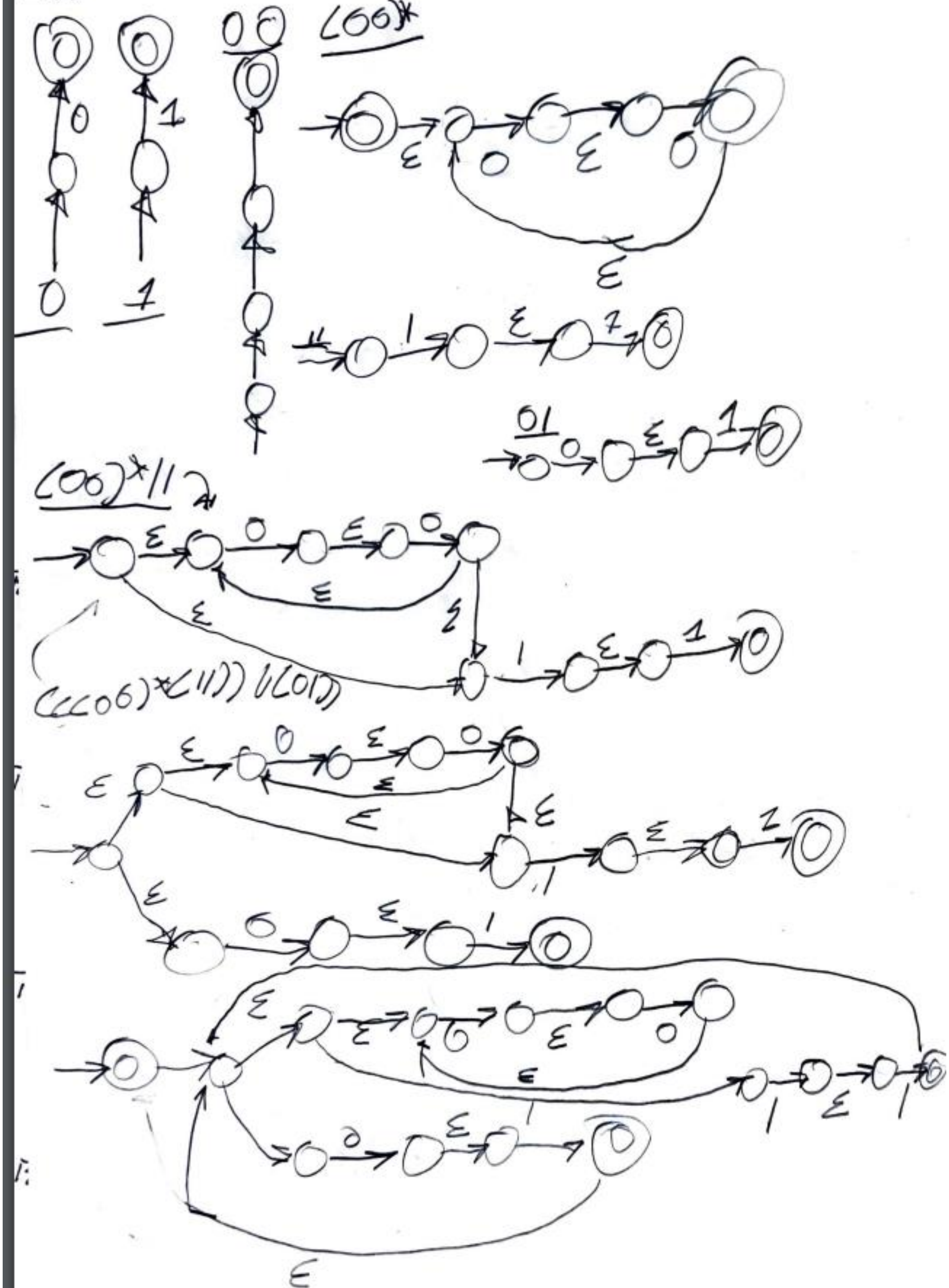
1.14a

$R = (0 \cup 1)^* 000 (0 \cup 1)^*$



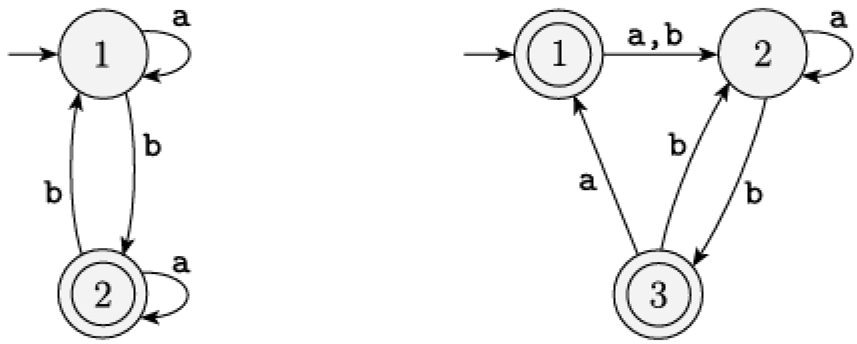
18h.

1.19b



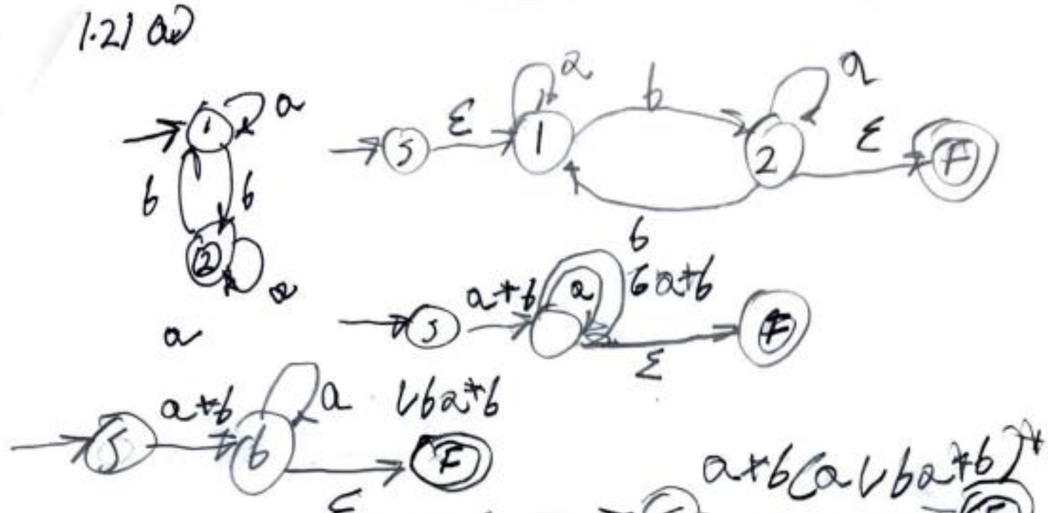
DFA to RE Algorithm: 1.21

1.21 Use the procedure described in Lemma 1.60 to convert the following finite automata to regular expressions.



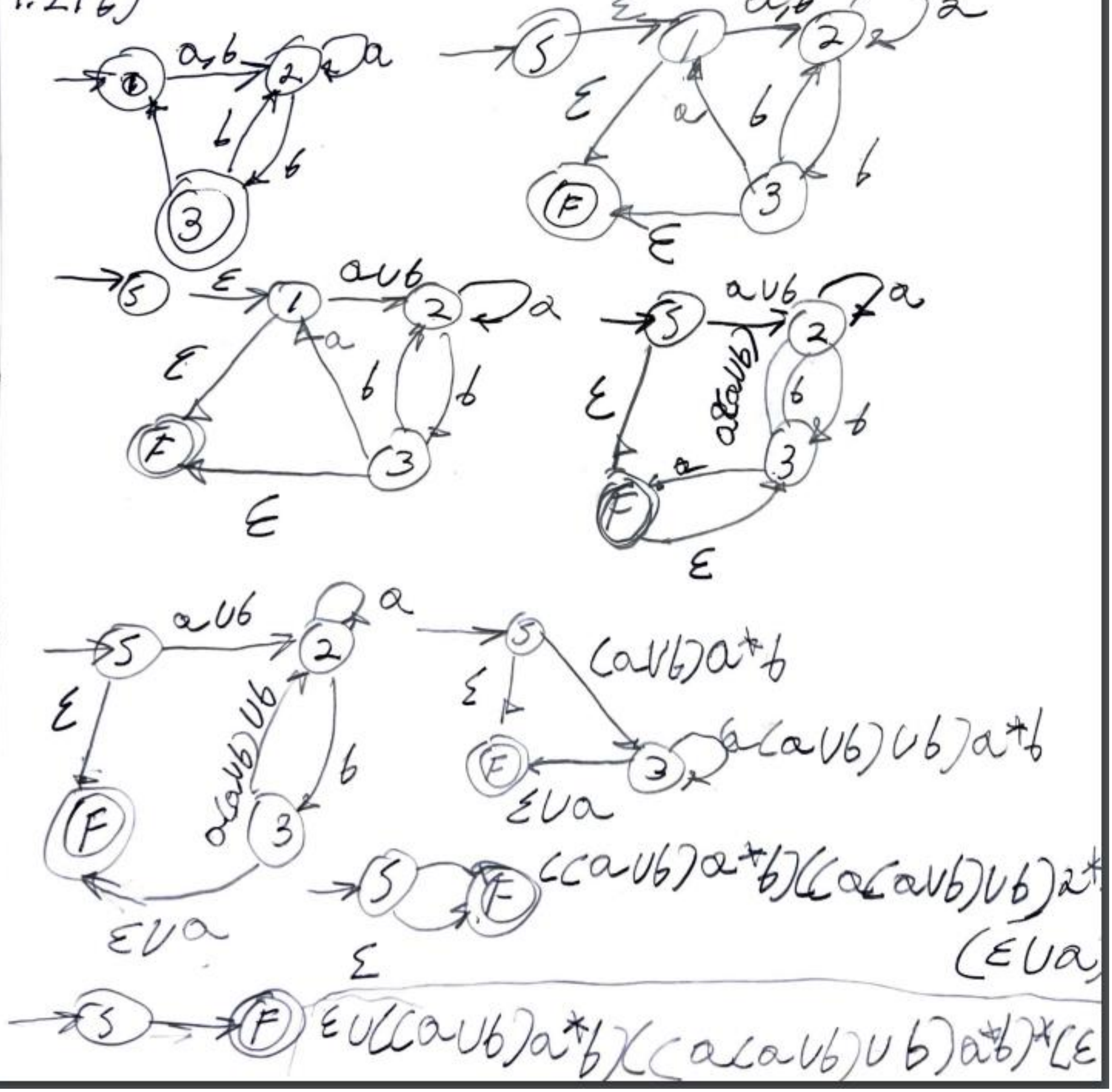
(a) (b)

1.21 a)



$\boxed{a^*b(a\cup ba^*b)^*}$

1.21 b)



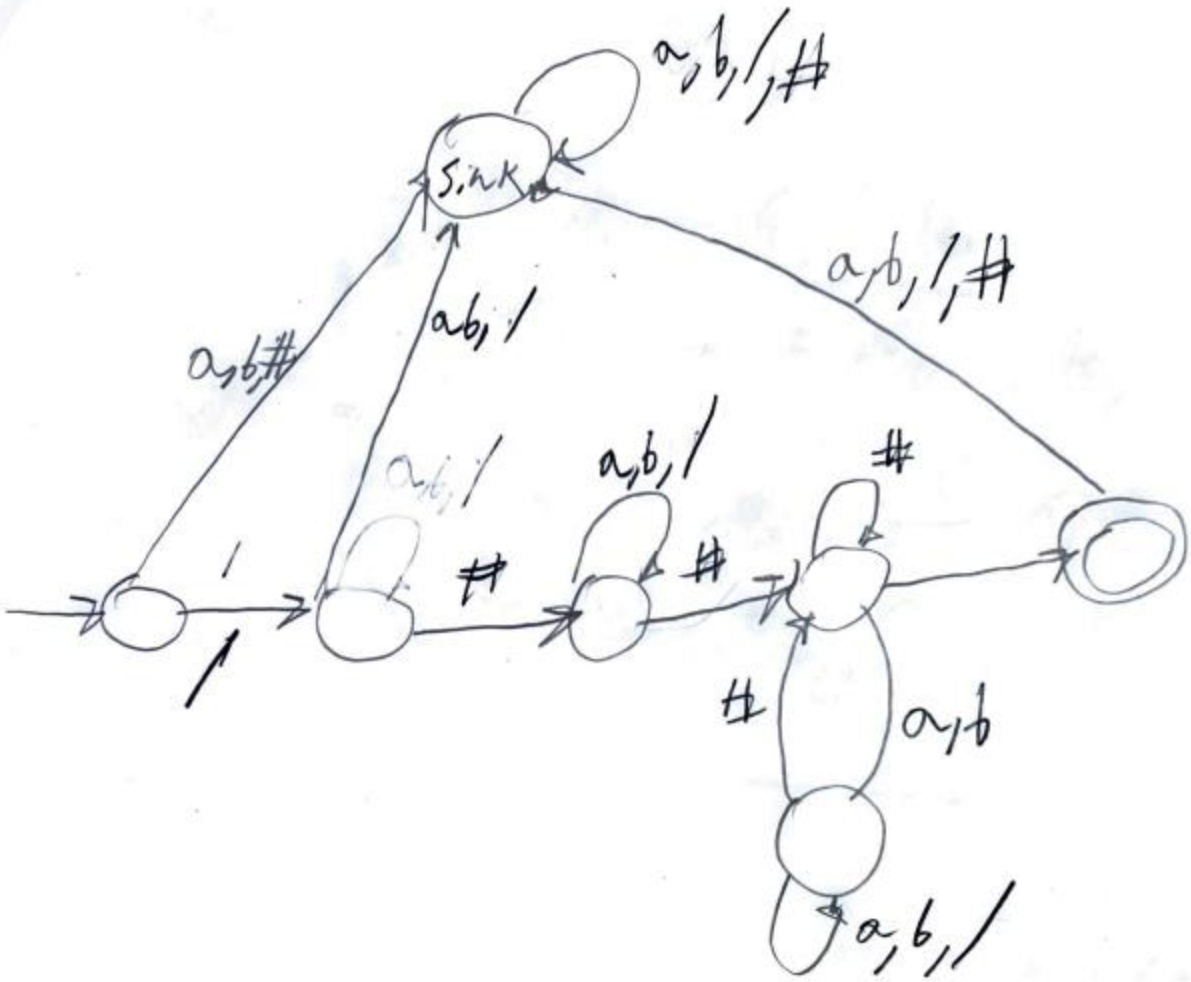
Regular Language: 1.22

1.22 In certain programming languages, comments appear between delimiters such as `/#` and `#/`. Let C be the language of all valid delimited comment strings. A member of C must begin with `/#` and end with `#/` but have no intervening `#/`. For simplicity, assume that the alphabet for C is $\Sigma = \{\texttt{a}, \texttt{b}, \texttt{/}, \texttt{\#}\}$.

a. Give a DFA that recognizes C.

b. Give a regular expression that generates C.

1.22



/#(a+b+/)*#(#+(a+b)(a+b+/)*#)*/

Pumping Lemma: 1.29a&b

1.29 Use the pumping lemma to show that the following languages are not regular.

[A]**a.** $A_1 = \{0^n 1^n 2^n \mid n \geq 0\}$

b. $A_2 = \{www \mid w \in \{a, b\}^*\}$

[A] [redacted]

1.29A

Pumping length = PL, str = string, Reg = Regular, Lang = Language

Switched from 1.22 to 1.29 a & b

1.29 a $A = \{0^n 1^n 2^n \mid n \geq 0\}$

PL: if A is Reg. Lang there's #P length, where S is any str. in A on length at least P, then S may be / into 3 pieces $S = xyz$, 1) for each $i \geq 0$, $xy^iz \in A$ 2) $|y| > 0$, 3) $|xy| \leq P$

Assume A_1 = reg. Lang

Let P = pumping length given by the pumping Lemma

~~Consider a str. $S = 0^P 1^P 2^P \in A$, $|S| \geq P$ so PL~~

Consider a Str $S = 0^P 1^P 2^P \in A$, $|S| \geq P$, so Pumping Lemma, take $S = 0^P 1^P 2^P = xyz$ such that $|xy| \leq P$, $|y| \geq 0$

Consider this

Let 001122 be a str that belongs to A_1 $S = 0^P 1^P 2^P$ = 001122. The P.L. of the str = 2. To satisfy condit of the pumping Lemma, x = 0, y = 0, z = 1122.

$S = 001122 = \underset{x}{00}\ \underset{y}{0}\ \underset{z}{1122}$

pump the middle such that xy^iz ($i \geq 0$), for $i = 2$, the y becomes 00. The str after pumping = 0001122

$S = (0)(0)^i(1122) = \underset{x}{0}\ \underset{y}{00}\ \underset{z}{1122}$ [i = 2]

The str 0001122 $\notin A_1$ b/c the str that is accepted by the lang should be = # of 0's, 1's, & 2's. ← contradiction! pumping Lemma violated.

Thus A_1 = NOT a Reg. Lang.

1.29B

1.29 b) $A_2 = \{www \mid w \in \{a,b\}^*\}$

A_2 is reg. Lang. Let P = pumping length given by the pumping lemma. Consider a str. $S = a^P b a^P b a^P b \in A$. By pumping lemma this str can be / int 3 pieces xyz such that $|xy| \le P$, $|y| > 0$ & $xy^iz \in A_2 \forall i \ge 0$ so $S = a^P b a^P b a^P b = xyz$

Let aabaabaab be the string that belongs to A_2. The pumping length of the str. = 2. To satisfy the conditions of the pumping lemma $x = a$, $y = a$, z = baabaab.

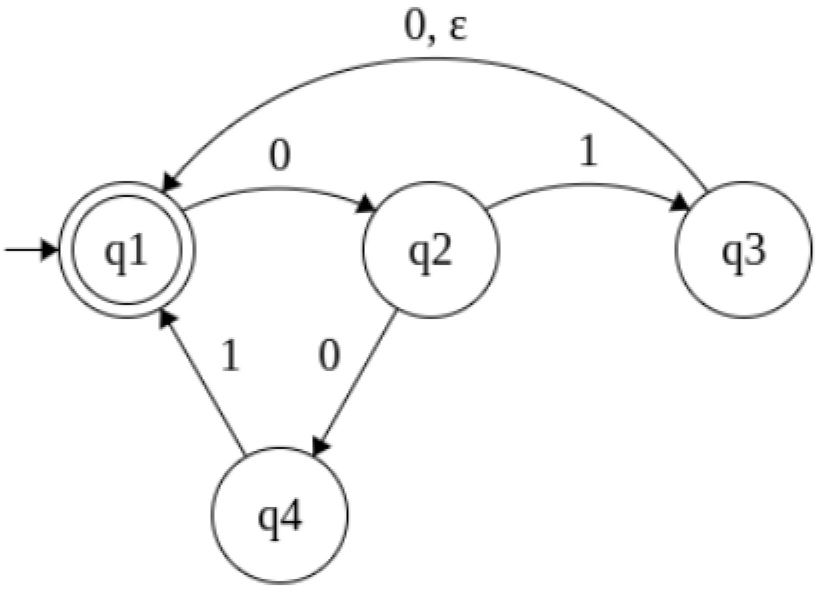
S = aabaabaab $= \underbrace{a}_{x} \underbrace{a}_{y} \underbrace{baabaab}_{z}$

Pump the middle part such that xy^iz ($i \ge 0$) for $i = 2$, the y becomes aa. The str. after pumping is aaabaabaab

$S = (a)(a)^i(baabaab) = \underbrace{a}_{x} \underbrace{aa}_{y} \underbrace{baabaab}_{z}$ [when $i = 2$]

Str. aaabaabaab $\notin A_2$. It's a contradiction. Pumping Lemma violated. Thus, A_2 is NOT a Reg. Lang.

NFA to DFA algorithm: 1.17b (Use the NFA given in class, which follows here)

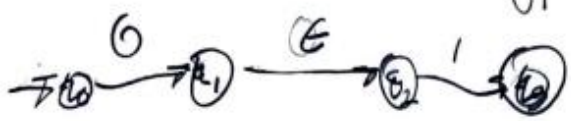


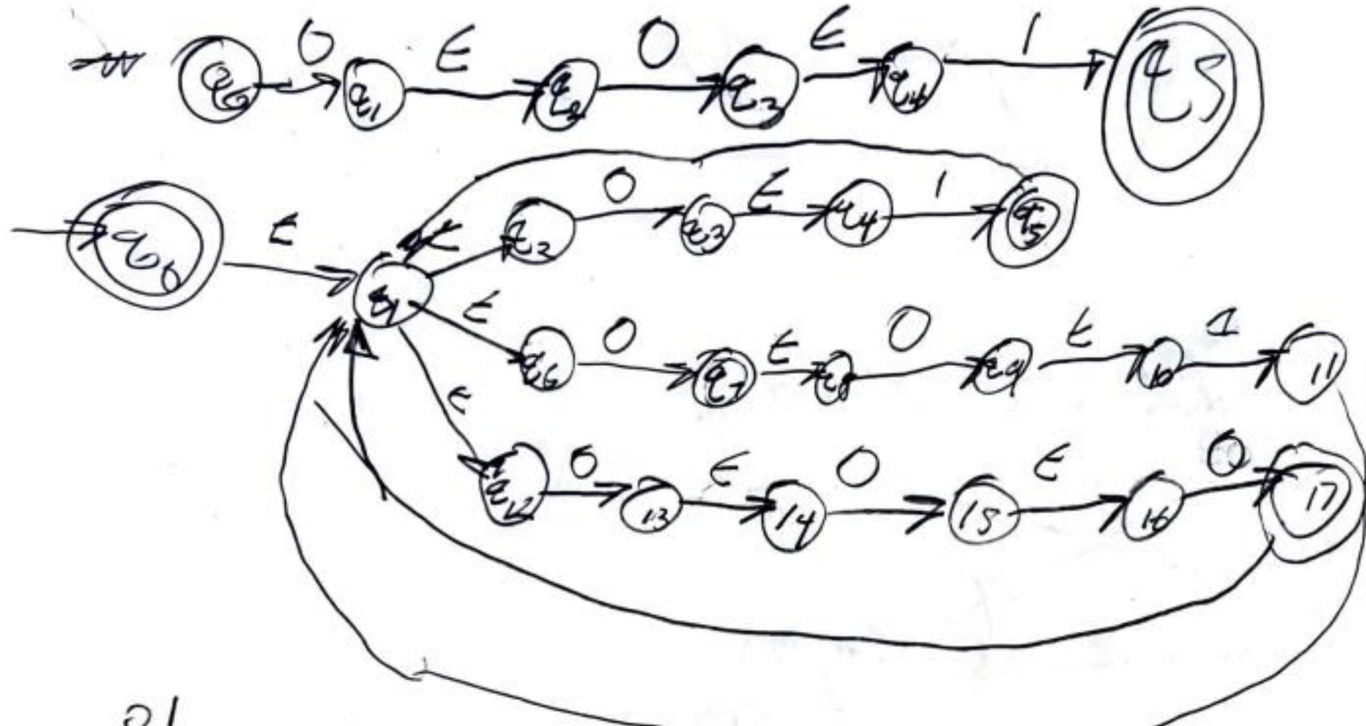
1.17 [redacted]

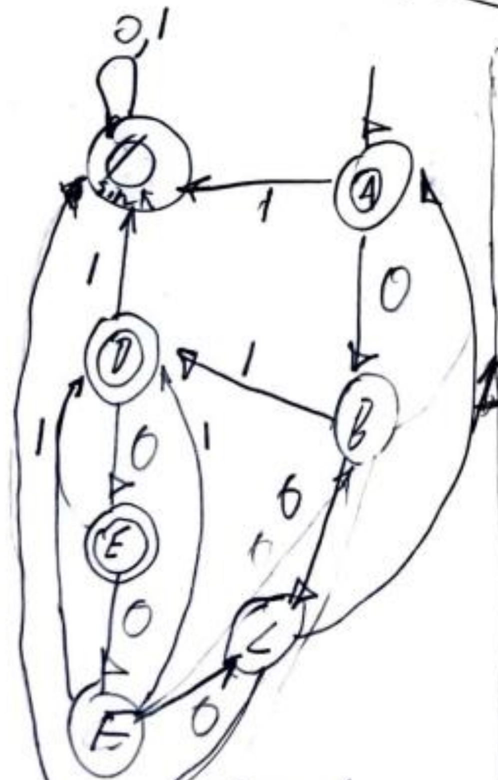
b. Convert this NFA to an equivalent DFA. Give only the portion of the DFA that is reachable from the start state.

1.17

01 $L=(01 \cup 001 \cup 010)^*$







DFA

I'm a little confuse so 1.17b & ~~do~~ next to NFA → DFA for the graph below? I think?

1.17b ust The NFA Given in Class